Where You Grow Up Matters

Predicting Adult Income for Low-Income Children in the DC Region

Capstone Two Final Report | Springboard Data Science Career Track | Prepared by Brittany | November 2026

Summary

Children who grow up in similar low-income families end up in very different places as adults depending on the neighborhood they grew up in. Across 735 census tracts in five DC-area counties, the average adult income rank of these children is **0.37 in DC, 0.40 in Maryland, and 0.51 in Virginia** (0 is the bottom of the national income distribution and 1 is the top). A simple model built from neighborhood characteristics predicts these outcomes well: on 147 tracts it had never seen, it explained 83% of the variation in income rank and was off by about **2.8 percentile points** on average. The strongest signals were the share of Black residents, household income, the share of Asian residents, and the share of foreign-born residents. These reflect how opportunity is distributed across places, shaped by a long history of segregation and unequal investment, **not** anything about the people who live there. The report ends with three recommendations for a foundation or community development agency.

1. The problem and who it is for

A child's chances of climbing the income ladder depend heavily on where they grow up. For a foundation deciding where to direct grants, or a community development agency deciding where to build housing, services, or programs, the practical question is: **which neighborhood characteristics are most closely tied to better adult outcomes for children from low-income families, and can they be used to find the places that need attention first?**

This project answers that question for the DC metro area: the District of Columbia, Prince George's County (MD), and Arlington, Fairfax, and Alexandria (VA). The two intended readers are a DC-area foundation and a community development agency, so the report ends with concrete recommendations for each.

1.1 What is measured

The outcome is the **adult income rank**: the average household income rank (0 to 1) at ages 31 to 37 for children who grew up in families at the 25th percentile of the national income distribution. A value of 0.45 means a typical child from such a family reaches about the 45th percentile of household income as an adult. The predictors describe each neighborhood (census tract) in 2000 to 2015: income, poverty, education, employment, family structure, race and ethnicity, commuting, density, and jobs nearby.

2. Data and approach

2.1 Data

Two public files were combined: the **Opportunity Atlas** tract outcomes (what happened to children who grew up in each tract, born 1978 to 1983) and a tract-level file of neighborhood characteristics from

Opportunity Insights. Both were national files of about 73,000 tracts. They were filtered to the five counties and joined on an 11-digit tract identifier, leaving 747 matched tracts.

2.2 Steps taken

- **Data wrangling.** Dropped 3 tracts with no outcome, and 3 columns that were mostly empty (wage growth 41.8% missing, two-bedroom rent 35.2%, job growth 24.0%). Filled the few remaining small gaps with medians. Found no duplicate tracts. Dropped 9 tracts with fewer than 20 children because their estimates are very noisy. The final table has 735 tracts. (An earlier download of a "top 1%" file turned out to be a copy of another file, and was replaced.)
- **Exploratory analysis.** Looked at every predictor, tested how strongly each relates to the outcome (Pearson and Spearman correlations with formal hypotheses), compared the three states with ANOVA, Kruskal-Wallis, and Tukey tests, and found four pairs of overlapping predictors. Dropped one from each pair, and dropped the third-grade math score because it is a county-level value that cannot separate tracts within a county. Log-transformed the two very skewed density columns.
- **Pre-processing.** Turned state into 0/1 dummy columns (DC as the baseline), split the data 80/20 into 588 training and 147 testing tracts (stratified by state so each set has the same state mix), and standardized the 15 continuous predictors using the training data only so no information leaks from the test set.
- **Modeling.** Compared a baseline (always guess the average) with three kinds of models: linear regression, ridge and lasso regression (linear models that handle overlapping predictors), and a tuned random forest. Because the outcome is continuous, the models were judged with R^2, mean absolute error (MAE), and root mean squared error (RMSE), using 5-fold cross-validation on the training data only.

3. Findings

3.1 Where a child grows up matters, and the DC region is highly uneven

Average adult income rank is 0.51 for tracts in Northern Virginia, 0.40 in Prince George's County, and 0.37 in DC (Figure 1). The spread within each area is also wide: many DC tracts are as high as the Virginia median, and many Virginia tracts are as low as DC's. In the exploratory analysis, state alone accounted for about 54% of the variation in the outcome, and all three pairwise differences between states were statistically significant.

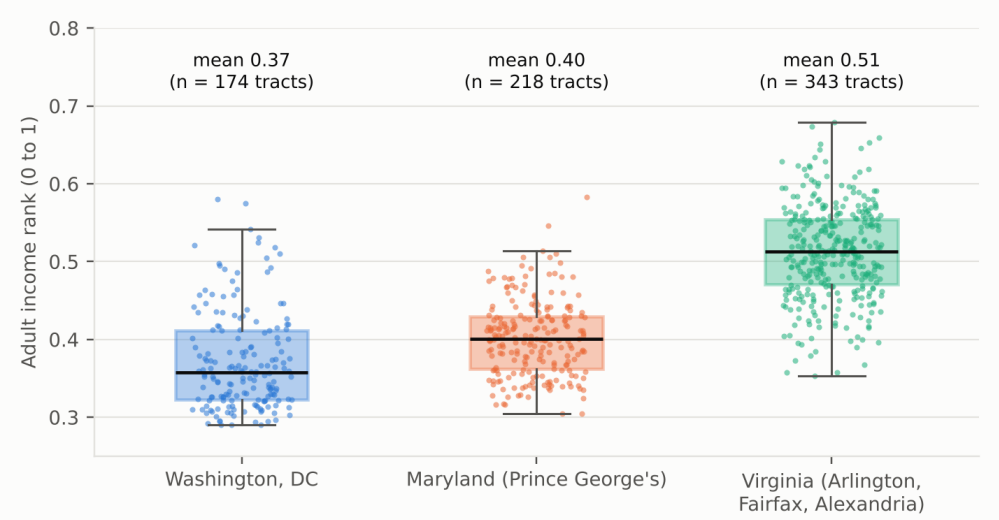


Figure 1. Adult income rank for children raised in low-income families, by state. Each dot is one census tract; boxes show the middle half of tracts and the black line is the median.

3.2 Education, income, family structure, and neighborhood composition track outcomes most closely

The relationships with the outcome are strong for several predictors (Figure 2). Tracts with higher past household income, more college-educated adults, and higher employment have higher income ranks for low-income children. Tracts with more single-parent households, higher poverty, and a higher share of Black residents have lower ones. Job density showed almost no relationship (correlation of -0.05). All of these are associations at the neighborhood level.

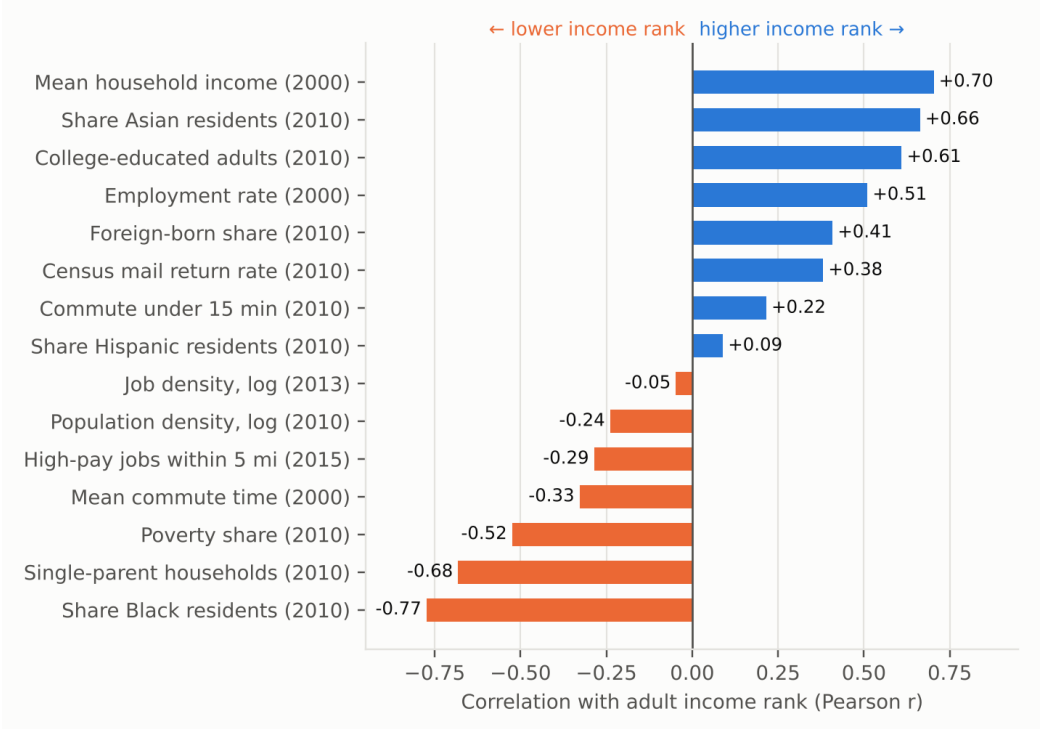


Figure 2. How strongly each neighborhood characteristic is related to adult income rank across 735 tracts. Bars show Pearson correlations (-1 to +1). Correlation is not causation.

3.3 A simple model predicts the outcome well

Table 1 compares the models. The random forest scored highest in cross-validation, but only by 0.012 R^2 points, which is smaller than the uncertainty in the comparison (standard error 0.014). Using the one-standard-error rule (choose the simplest model whose score is within one standard error of the best), **linear regression was chosen as the final model**. It is simpler to explain, cheaper to run, and easier to maintain, and it gives up almost nothing in accuracy.

Model	CV R^2	CV std. error	CV MAE (pct. points)	Test R^2	Test MAE (pct. points)
Baseline (guess the average)	-0.029	0.010	7.1	-0.001	7.6
Linear regression (final model)	0.745	0.017	3.1	0.833	2.8
Ridge regression	0.747	0.015	3.0	0.826	2.9
Lasso regression	0.749	0.015	3.0	0.829	2.9
Random forest (tuned)	0.757	0.014	2.9	0.846	2.7

Table 1. Model comparison. CV = 5-fold cross-validation on the 588 training tracts (the basis for choosing the model). Test = 147 held-out tracts, used once at the end. MAE is the average miss in percentile points of income rank.

On the 147 test tracts, the final model reaches $R^2 = 0.83$ and an average miss of 2.8 percentile points, compared with 7.1 points for simply guessing the average (Figure 3). The test score is higher than the cross-validated score (0.74), which is plausible by chance with only 147 test tracts. The more conservative cross-validated figure is the safer one to quote. The model does not over- or under-predict any state on average, but its errors are somewhat larger for tracts with higher predicted ranks (3.2 versus 2.5 percentile points), and largest in Virginia (3.2 points, versus 2.7 in DC and 2.4 in Maryland).

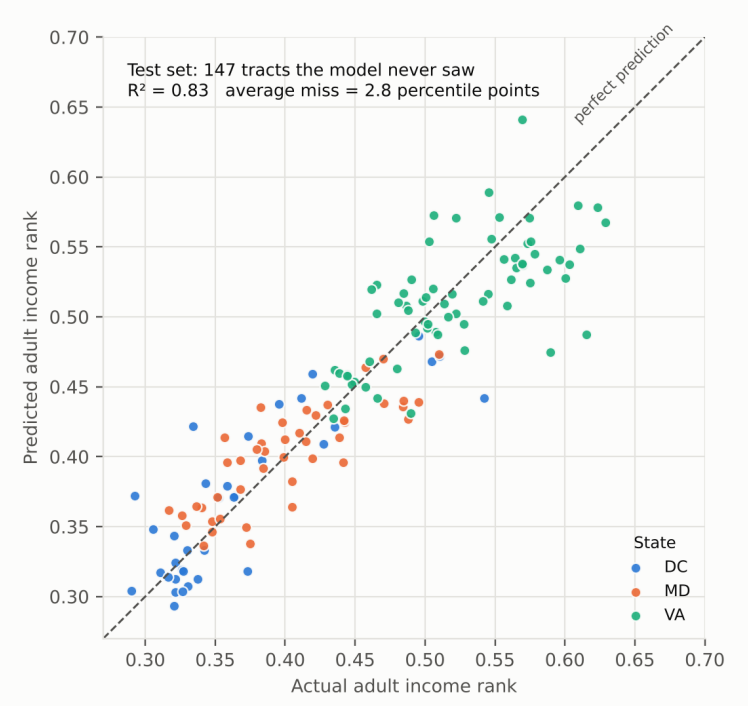


Figure 3. Final model: predicted versus actual income rank on the test set. Points near the dashed line are accurate predictions.

3.4 What the final model relies on

Figure 4 shows what the final model uses. Share of Black residents carries the largest weight (a drop of about 3.5 percentile points per standard deviation), followed by Virginia versus DC (+2.0), mean household income in 2000 (+1.9), share of Asian residents (+1.4), and foreign-born share (+1.3). Two cautions apply. First, **overlapping predictors share credit**: single-parent households and poverty are strongly related to the outcome on their own, but their weights here are small because they overlap with income and share Black (the largest variance inflation factor was 11.4, and six predictors were above the usual cutoff of 5). The model predicts well, but individual weights should not be read as separate effects. Second, the state columns add little once neighborhood characteristics are included: test R^2 is 0.824 without them and 0.833 with them, so most of the gap between states is accounted for by neighborhood conditions.

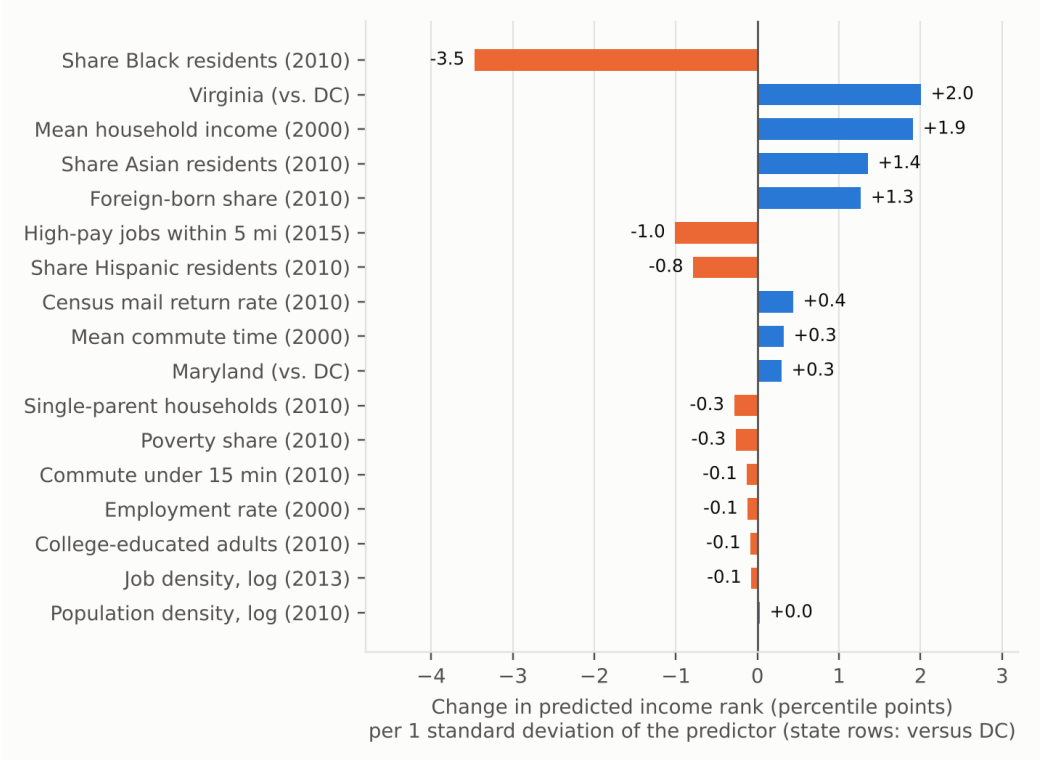


Figure 4. Weights in the final linear model. Predictors were standardized, so each bar shows the change in predicted income rank (in percentile points) for a one-standard-deviation increase in that predictor, holding the others fixed.

4. Recommendations

These recommendations are written for a DC-area foundation and a community development agency. They follow from associations in the data, so each includes a way to check it locally before committing large resources.

Recommendation 1: Use the model as a screening tool to decide where to look first, not as a verdict on any family

Rank census tracts by predicted adult income rank and flag the lowest for closer review. On the test set, the model placed **33 of the 37 tracts (89%)** that were actually in the bottom quarter of outcomes in its own bottom quarter. Use it to focus on-the-ground assessment, listening sessions, and grant

outreach. It should not be used to decide eligibility for any individual or household, and it is less precise for higher-ranked tracts.

Recommendation 2: Direct place-based investment to neighborhoods with low predicted outcomes and the conditions linked to them

The conditions most closely tied to lower outcomes were low household income, lower employment, fewer college-educated adults, and higher poverty (correlations of +0.70, +0.51, +0.61, and -0.52 with income rank). Prioritize investments such as workforce programs, adult education, small-business support, and affordable housing in tracts that score low on these measures. The strong racial composition signal should be read as a marker of **historic segregation and disinvestment in places**. Use it to find neighborhoods that have been underinvested in, never to target or exclude people by race. Start with pilots in a few tracts and measure what changes, since these findings show association and not cause.

Recommendation 3: Invest in better local data and evaluation before scaling up

The model leaves about 17% of the variation unexplained, its errors are larger for higher-ranked tracts, and the outcomes describe children born in 1978 to 1983 while the predictors describe 2000 to 2015. Before committing large sums, fund tract-level data on school quality, housing stability and cost, and transit access (the school measure available here was county-level, and rent and job growth were too incomplete to use), and track outcomes for the places and programs you fund. That makes future versions of the model more accurate and shows whether investments actually move outcomes.

5. Limitations and ideas for further research

5.1 Limitations

- **Association, not cause.** The data describe neighborhoods, not individual children. A strong predictor does not mean changing it would change a child's outcome.
- **Small sample.** 735 tracts, and a test set of 147, so test scores move around; cross-validation is the more conservative guide.
- **Noisy estimates for small tracts.** Tracts with fewer than 20 children were removed, but many remaining tracts have small counts. The outcome file also adds a small amount of random noise for privacy (some incarceration rates came out slightly negative as a result).
- **Timing gap.** Outcomes are for children born 1978 to 1983 (measured in 2014 to 2015); predictors describe 2000 to 2015. Neighborhoods have changed since, especially in DC.
- **Overlapping predictors.** Several predictors move together, so individual model weights are unstable even though predictions are good.

5.2 Ideas for further research

- Weight each tract by the number of children it contains, so reliable tracts count more.
- Add school quality, housing, and transit measures, and test whether they explain the part of the outcome left over.
- Study outcomes for specific groups (by race and gender within tracts), which the Opportunity Atlas also provides.

- Try other model types such as gradient boosting, and compare against the linear model with the same one-standard-error rule.
- Extend the analysis to other metro areas to see whether the same predictors matter.

Appendix: submission files

- **Notebooks (Google Colab / Jupyter):** Data Wrangling, Exploratory Data Analysis, Pre-processing and Training Data, Modeling.
- **model_metrics.txt:** final model features, parameters, hyperparameters, and performance metrics.
- **README.md:** project summary and guide to the repository.
- **Data sources:** Opportunity Atlas tract outcomes and Opportunity Insights tract covariates (opportunityatlas.org and opportunityinsights.org).